

Functional Requirements Document (FRD)

IntelliPlant AI

AI-Powered Industrial Knowledge Intelligence Platform

Document Version: 1.0

Document Type: Functional Requirements Document (FRD)

Prepared For: ET AI Hackathon 2026

Prepared By: Team IntelliPlant AI

1. Purpose

The Functional Requirements Document (FRD) defines all functional capabilities, workflows, business processes, and system behaviors required for IntelliPlant AI.

This document specifies what the system should do and how users interact with the system.

2. Scope

The system shall provide:

- Document ingestion
 - OCR processing
 - Entity extraction
 - Knowledge graph generation
 - Hybrid search
 - Conversational AI
 - Root Cause Analysis (RCA)
 - Compliance Intelligence
 - Lessons Learned Intelligence
 - Multi-Agent orchestration
 - User management
 - Audit logs
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3. Actors

Administrator

Manages users, roles, and system settings.

Maintenance Engineer

Analyzes failures and maintenance records.

Reliability Engineer

Performs root cause analysis.

Quality Engineer

Checks standards and compliance.

Plant Manager

Monitors operations and reports.

Field Technician

Accesses SOPs and manuals.

4. Functional Modules

Module ID	Module
M-01	Authentication
M-02	User Management
M-03	Document Ingestion
M-04	OCR Engine
M-05	Metadata Extraction

Module ID	Module
M-06	Entity Extraction
M-07	Knowledge Graph
M-08	Embedding Pipeline
M-09	Hybrid Search
M-10	Conversational AI
M-11	Root Cause Analysis
M-12	Compliance Intelligence
M-13	Lessons Learned Engine
M-14	Multi-Agent Supervisor
M-15	Dashboard
M-16	Audit Logs

5. Functional Requirements

M-01 Authentication Module

FR-001 User Login

Description

Users shall authenticate using email and password.

Inputs

- Email
- Password

Output

JWT Token

Priority

Critical

FR-002 Logout

Users shall securely terminate sessions.

Priority: High

FR-003 Password Reset

Users shall reset passwords through email verification.

Priority: Medium

M-02 User Management Module

FR-004 Create User

Administrator shall create users.

Fields:

- Name
- Email
- Role

Priority:

High

FR-005 Assign Roles

Supported roles:

- Admin
- Manager
- Engineer
- Technician

Priority:

High

FR-006 Disable User

Admin shall disable accounts.

Priority:

Medium

M-03 Document Ingestion Module

FR-007 Upload Documents

Supported Formats

- PDF
- DOCX
- XLSX
- CSV
- JPG
- PNG
- TXT

Priority:

Critical

FR-008 Bulk Upload

System shall support batch uploads.

Priority:

High

FR-009 Metadata Capture

Store:

- Filename
- Upload date

- Owner
- Document type

Priority:

High

M-04 OCR Module

FR-010 OCR Processing

Extract text from:

- Images
- Scanned PDFs
- P&ID drawings

Priority:

Critical

FR-011 Table Extraction

Extract structured tables.

Priority:

Medium

FR-012 Image Caption Extraction

Extract text from diagrams.

Priority:

Medium

M-05 Metadata Extraction Module

FR-013 Document Classification

Classify documents into:

- Manual
- SOP
- Incident Report
- Inspection Report
- Work Order

Priority:

High

FR-014 Date Extraction

Extract:

- Inspection dates
- Maintenance dates

Priority:

High

M-06 Entity Extraction Module

FR-015 Equipment Extraction

Identify:

- Pumps
- Motors
- Valves
- Boilers

Priority:

Critical

FR-016 Personnel Extraction

Extract:

- Operators
- Engineers

Priority:

Medium

FR-017 Regulation Extraction

Identify:

- OISD
- Factory Act
- ISO Standards

Priority:

High

FR-018 Failure Mode Extraction

Extract:

- Bearing failure
- Leakage
- Overheating

Priority:

Critical

M-07 Knowledge Graph Module

FR-019 Node Creation

Create nodes:

- Equipment

- Manual
- Incident
- Operator
- Regulation

Priority:

Critical

FR-020 Relationship Generation

Relationships:

CONNECTED_TO

CAUSES

FAILED_IN

DESCRIBED_BY

SIMILAR_TO

REQUIRES

Priority:

Critical

FR-021 Graph Querying

Support Cypher-based search.

Priority:

Medium

M-08 Embedding Pipeline

FR-022 Chunk Documents

Split documents semantically.

Priority:

Critical

FR-023 Generate Embeddings

Generate vector embeddings.

Priority:

Critical

FR-024 Store Vectors

Store inside Qdrant.

Priority:

Critical

M-09 Hybrid Search Module

FR-025 Semantic Search

Natural language search.

Priority:

Critical

FR-026 BM25 Search

Keyword retrieval.

Priority:

High

FR-027 Graph Search

Knowledge graph traversal.

Priority:

High

FR-028 Hybrid Retrieval

Combine:

- Vector search
- BM25
- Graph search

Priority:

Critical

M-10 Conversational AI Module

FR-029 Ask Questions

Users shall query using natural language.

Priority:

Critical

FR-030 Citation Support

Answers shall contain sources.

Priority:

Critical

FR-031 Confidence Score

Provide answer confidence.

Priority:

High

FR-032 Follow-up Questions

Support conversation history.

Priority:

High

M-11 Root Cause Analysis Module

FR-033 Similar Incident Search

Retrieve historical failures.

Priority:

Critical

FR-034 Failure Pattern Analysis

Analyze recurring failures.

Priority:

Critical

FR-035 Root Cause Recommendation

Generate probable causes.

Priority:

Critical

FR-036 Maintenance Suggestions

Recommend corrective actions.

Priority:

High

M-12 Compliance Intelligence Module

FR-037 Compliance Mapping

Map documents against:

- Factory Act
- OISD
- ISO

Priority:

Critical

FR-038 Compliance Gap Detection

Identify missing requirements.

Priority:

Critical

FR-039 Audit Evidence Generation

Generate reports.

Priority:

High

M-13 Lessons Learned Engine

FR-040 Incident Analysis

Analyze:

- Near misses
- Failures
- Audit findings

Priority:

Critical

FR-041 Pattern Detection

Find repeated problems.

Priority:

High

FR-042 Recommendation Generation

Provide preventive measures.

Priority:

High

M-14 Multi-Agent System

FR-043 Supervisor Agent

Route tasks to agents.

Priority:

Critical

FR-044 Expert Copilot Agent

Answer engineering questions.

Priority:

Critical

FR-045 RCA Agent

Perform failure analysis.

Priority:

Critical

FR-046 Compliance Agent

Check regulations.

Priority:

Critical

FR-047 Lessons Agent

Analyze incidents.

Priority:

High

M-15 Dashboard

FR-048 Document Dashboard

Display uploaded documents.

Priority:

High

FR-049 Knowledge Graph Visualization

Display graph relationships.

Priority:

Medium

FR-050 Analytics Dashboard

Show:

- Total documents
- Incidents
- Equipment

Priority:

Medium

M-16 Audit Module

FR-051 Query Logging

Track user queries.

Priority:

High

FR-052 User Activity Logs

Track system actions.

Priority:

High

FR-053 Document History

Maintain version history.

Priority:

Medium

6. Business Rules

BR-01

Only authenticated users can access the platform.

BR-02

Every answer must contain citations.

BR-03

Knowledge graph updates automatically.

BR-04

Documents are immutable after ingestion.

BR-05

Deleted documents are archived.

BR-06

Admin has full privileges.

7. Functional Dependencies

Document Upload

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OCR

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Metadata Extraction

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Entity Extraction

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Chunking

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Embedding Generation

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Vector Storage

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Knowledge Graph

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Hybrid Search

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Agents

↓

LLM

↓

Response Generation

8. Error Handling Requirements

OCR Failure

Retry mechanism.

LLM Failure

Fallback response.

Embedding Failure

Queue processing.

Graph Failure

Partial retrieval mode.

Database Failure

Automatic recovery.

9. Acceptance Criteria

System shall:

- ✓ Upload documents
 - ✓ Extract text
 - ✓ Build knowledge graph
 - ✓ Generate embeddings
 - ✓ Support semantic search
 - ✓ Provide citations
 - ✓ Perform RCA
 - ✓ Detect compliance gaps
 - ✓ Generate lessons learned
 - ✓ Support multi-agent workflows
 - ✓ Maintain audit logs
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10. Conclusion

The Functional Requirements Document defines all business functions and user interactions necessary for IntelliPlant AI. These requirements establish the functional foundation for architecture design, APIs, database schemas, AI workflows, and implementation.